

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		B iodine C ice E sodium chloride F caesium chloride award (2) for all four correct award (1) for any two correct	2			2		
		(ii)		ice iodine cadmium graphite award (2) for all four correct award (1) for any two correct		2		2		
		(iii)		Van der Waals / induced dipole-induced dipole accept London forces / dispersion forces do not accept dipole-dipole	1			1		
		(iv)		graphite and cadmium both needed		1		1		
		(v)		weak forces between the layers are easily broken (1) layers can slide over each other / be removed (and mark the paper) (1)	2			2		

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					AO1	AO2	AO3	Total	Maths	Prac
		(vi)		metals consist of a regular arrangement / lattice of metal cations (1) surrounded by a 'sea' of delocalised electrons (1) there are electrostatic forces of attraction between them (1) first two marks can be obtained from suitable diagram	3			3		
	(b)	(i)	I	white precipitate	1			1		1
			II	colourless solution / no observable change	1			1		1
		(ii)		$\text{Cd}^{2+}(\text{aq}) + 2\text{OH}^{-}(\text{aq}) \rightarrow \text{Cd}(\text{OH})_2(\text{s})$ accept Mg^{2+} instead of Cd^{2+} accept $\text{Cd}^{2+}(\text{aq}) + \text{SO}_4^{2-}(\text{aq}) \rightarrow \text{CdSO}_4(\text{aq})$ accept incorrect state symbols for products if consistent with observations in (b)(i)		1		1		
				Question 8 total	10	4	0	14	0	2